

General Characteristics

41. Which is not true for transition elements
 (a) They are all metals
 (b) They show variable valency
 (c) They form coloured ions
 (d) They do not form co-ordinate compounds
42. The main reason for not using a mercury electrolytic cell in NaOH manufacture is that
 (a) Hg is toxic
 (b) Hg is a liquid
 (c) Hg has a high vapour pressure
 (d) Hg is a good conductor of electricity
43. Lanthanum is grouped with f -block elements because
 (a) It has partially filled f -orbitals
 (b) It is just before Ce in the periodic table
 (c) It has both partially filled f and d -orbitals
 (d) The properties of Lanthanum are very similar to the elements of $4f$ block
44. The element having electronic configuration belongs to $ns^2(n-1)d^{1-10}(n-2)f^{1-14}$
 (a) s -block (b) p -block
 (c) d -block (d) f -block
45. Variable valency is shown by
 (a) Typical elements (b) Normal elements
 (c) Transition elements (d) None of these
46. Which ion is not coloured
 (a) Cr^{3+} (b) Co^{2+}
 (c) Cr^{2+} (d) Cu^+
47. The number of unpaired electrons in ferrous ion is
 (a) 5 (b) 4
 (c) 3 (d) 2
48. Fe , Co and Ni have valuable catalytic properties in process involving
 (a) Organic compound
 (b) Oxidation
 (c) Hydrogenation
 (d) Compounds of hydrogen
49. Which of the following statement is not correct
 (a) Metals contribute their valency electrons to the common sea of electrons
 (b) Metals have high co-ordination number
 (c) Metals tend to adopt closely packed structures
 (d) Metals have high lattice energy
50. Zinc, cadmium and mercury show the properties of
 (a) Typical elements
 (b) Normal elements
 (c) Transitional elements



- (d) Rare elements
51. Iron is
(a) A normal element
(b) A typical element
(c) A transitional element
(d) An inert element
52. Platinum, palladium, iridium etc., are called noble metals because
(a) Alfred Noble discovered them
(b) They are inert towards many common reagents
(c) They are shining lustrous and pleasing to look at
(d) They are found in active state
53. Which of the following statement is not true about *Mohr's salt*
(a) It decolourises $KMnO_4$
(b) It is a primary standard
(c) It is a double salt
(d) Oxidation state of iron is +3 in this salt
54. Which one of the following statement is true for transition elements
(a) They exhibit diamagnetism
(b) They exhibit inert pair effect
(c) They do not form alloys
(d) They show variable oxidation states
55. The valence shell electronic configuration of Cr^{2+} ion is
(a) $4s^0 3d^4$
(b) $4s^2 3d^2$
(c) $4s^2 3d^0$
(d) $3p^6 4s^2$
56. The hardness of *Cr* is due to and metallic lusture is due to
(a) Covalent bond, metallic bond
(b) Covalent bond, hydrogen bond
(c) Metallic bond, covalent bond
(d) Metallic bond, hydrogen bond
57. In the first transition series, the highest B.P. and M.P. is of
(a) *Cr*
(b) *V*
(c) *Ni*
(d) *Fe*
58. In the following transition elements, the lowest M.P. and B.P. is exhibited by
(a) *Cr*
(b) *Hg*
(c) *Cu*
(d) *Au*
59. In the following members of transition elements, the lowest ionization energy is of
(a) *Ti*
(b) *Sc*
(c) *V*
(d) *Mn*
60. Which of the following has second ionisation potential less than expected
(a) *Cr*
(b) *Zn*
(c) *V*
(d) *Mo*

